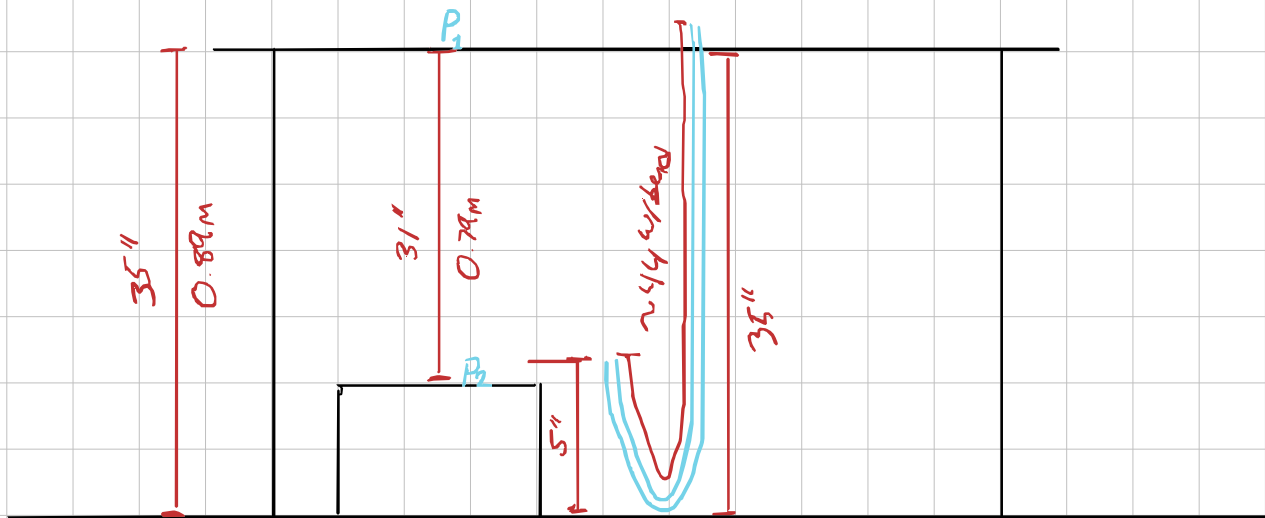


Counter to floor / basin



P₁

$$\frac{V_1^2}{2} + g z_1 + \frac{P}{\rho}$$

P₂

$$\frac{V_2^2}{2} + g z_2 + \frac{P}{\rho}$$

Pressure is the same at the outlet and the reservoir

V : Velocity of water at this point
 z : height of this point relative to outlet.

For the rhs, z_2 is zero, so the term simplifies. The same with V_1 , which is essentially zero vs V_2

$$\cancel{\frac{V_1^2}{2}} + g z_1 = \cancel{\frac{V_2^2}{2}} + g z_2$$

$$g z_1 = \frac{V_2^2}{2}$$

Get z_1 by itself.

$$z_1 = \frac{V_2^2}{2g}$$

Add the losses to the left side. First we cover head loss due to friction in the tubing h_f .

$$h_f = f \frac{L}{D} \frac{v^2}{2g} \left(\frac{m^2/s^2}{m/s^2} = m \right)$$

h_f is expressed as an equivalent height of water in meters.

f : Darcy Friction factor. Dimensionless. How strongly the tube resists flow.

L : length of the tubing

D : diameter of the inside of the tube

v : velocity of the water through the tubing section

g : gravitational acceleration

This gives us:

$$z_1 - h_f = \frac{v_2^2}{2g}$$

Or:

$$z_1 = \frac{v_2^2}{2g} + \underbrace{f \frac{L}{D} \frac{v_2^2}{2g}}_{\text{Tubing loss}} \quad \left(\text{estimate } f=0.04 \right)$$

Next is Coupler loss:

$$h_k = K \frac{v_2^2}{2g} \quad K = \frac{11}{50} \text{ for this coupler}$$

We get:

$$z_1 = \frac{v_2^2}{2g} + f \frac{L}{D} \frac{v_2^2}{2g} + \frac{11}{50} \frac{v_2^2}{2g}$$

$$z_1 = \frac{V_2^2}{2g} + f \frac{L}{D} \frac{V_2^2}{2g} + 11 \frac{V_2^2}{2g}$$

$$f = 0.04$$

$$L = \text{Tube length} = 43'' = 1.09 \text{ m}$$

$$D = 1D = 0.125'' = 0.003175 \text{ m}$$

SUBSTITUTE single appearance constants

$$z_1 = \frac{V_2^2}{2g} + 0.04(343.307) \frac{V_2^2}{2g} + 11 \frac{V_2^2}{2g}$$

$$z_1 = \frac{V_2^2}{2g} (1 + 13.73 + 11)$$

$$z_1 = 0.79 \text{ m}$$

$$2g(z_1) = (25.73 \left(\frac{V_2^2}{2g} \right)) 2g$$

$$\frac{2gz_1}{25.73} = \frac{25.73 V_2^2}{25.73}$$

$$\frac{9.8(2)(0.79)}{25.73} = V_2^2$$

$$\sqrt{0.602} = \sqrt{V_2^2}$$

$$0.776 \text{ m/s} = V_2$$



Assume 45° angle on tube outlet:

$$V_x = V_y = \sin(45) (0.776 \text{ m/s}) = (0.7071)(0.776) = 0.549 \text{ m/s}$$

Time in the air:

$$t = \frac{2V_y}{g} = \frac{2(0.549)}{9.81} = 0.112 \text{ s}$$

X Travel:

$$X = V_x t = 0.549(0.112) = 0.0615 \text{ m} = 6.2 \text{ cm} = 2.4 \text{ in}$$